Spencer Presley

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EXPERIENCE

CrunchAtlas — Software Engineer | Mar 2025 – Present

Local-first, air-gapped AI security platform for critical infrastructure. Sole author of the systems below.

- Created the company's agent platform from scratch. A shared agent harness supplies model construction, retries, structured-output repair, context management, tool guardrails, per-task budgets, and typed lifecycle events; each agent adds only its domain prompt, tools, state, and graph.
- Kept long-running agents on task beyond a single context window. As context fills or the model repeats a tool call, the harness condenses the run to its objective, completed work, and next steps; oversized tool results move to searchable files the agent reads on demand instead of resending them with every model call.
- Prevented context overruns and truncated model turns from corrupting an agent run. Before every llama.cpp call, the harness renders and tokenizes the exact outgoing request and refuses any prompt, summary, or retry that won't fit. If generation hits its cap, it drops the partial turn before memory or tool execution and regenerates once.
- Gave understaffed critical-infrastructure teams on-demand packet forensics through the platform's most-used agent. It drives tshark, Zeek, and foremost from an IDS alert through protocol reconstruction, IOC extraction, file carving, and timeline analysis, then delivers a cited report and triage verdict for defenders to review.
- Made the platform investigate on its own: a background reconciler decides which AI work should exist and queues it with no analyst click, completing nearly 17,000 PCAP investigations in its first two months. Each dispatch commits independently behind a durable resume cursor, and a user's click promotes an already-queued run ahead of all background work.
- Shipped PurpleHaze, an autonomous penetration-testing product covering reconnaissance, exploitation, web, and credential testing. Its Metasploit tool keeps modules, options, and Meterpreter sessions alive across model turns and context compaction, so a full engagement ends in OWASP and NIST reports.
- Replaced fixed LLM concurrency with token-priced GPU admission. Five fixed 131k-token llama.cpp slots became one 655k-token KV pool that holds any job mix whose declared context fits, from 7 concurrent 128k-context agents to 160 4k-token jobs, sized byte-exact against llama.cpp's KV-cache allocation.
- Kept long-running GPU and sandbox jobs independently cancellable and returned in-flight work to a terminal state after crashes or hangs. The Postgres scheduler combines atomic `SELECT FOR UPDATE SKIP LOCKED` claims, interactive-first, per-customer fair-share ordering, per-run process isolation, heartbeats, and process-tree cancellation.

Horn Point Laboratory — Software Engineering Intern | May 2024 – Aug 2024

- Made an NSF-funded Chesapeake Bay salinity corpus directly searchable through a retrieval-augmented chatbot, built on FAISS/OpenAI retrieval and a streaming FastAPI/LangChain service; presented at MIT URTC.

PROJECTS

docq — *Python, SQLite/FTS5, Ollama*

- Created a cited retrieval engine that lets coding agents search a long reference instead of loading it into context. On a leakage-screened 266-case eval, hybrid BM25/vector retrieval reached 0.67 hit@1 and beat either channel alone; a skill-primed query-rewrite A/B raised it to 0.74 (p=0.0045).

BitNet b1.58 (from scratch) — *PyTorch, ternary-weight LM, byte-level, Android*

- Trained ternary-weight ({−1, 0, +1}) language models from scratch at three sizes up to 300M parameters, then deployed one to an Android phone and profiled on-device CPU, memory, and energy.

HONORS

- **1st Place / Best Hack — Bitcamp 2024**, UMD (largest US East Coast hackathon) — TermBook
- **Best Educational Hack & 2nd Overall — HackUMBC 2024** — Testif.AI

SKILLS

- **AI & agents:** Agent orchestration, tool calling, context engineering, guardrails, prompt-injection defense, model evaluation, RAG, LLM serving, LangChain, LangGraph, llama.cpp, llama-swap, Ollama, OpenAI API, PyTorch
- **Languages:** Python, Rust, TypeScript, SQL, Bash
- **Backend & infrastructure:** Django, FastAPI, PostgreSQL, Redis, Docker, Linux, AWS, Terraform, GitHub Actions

EDUCATION

Salisbury University — B.S. Computer Science, Minor in Mathematics | Aug 2021 – May 2025